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BE (6th Semester)

Examination, Nov.-Dec., 2013

Branch : IT

COMPUTER GRAPHICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part 'a' is compulsory and attempt any two questions from rest. All questions carry equal marks.

Q. 1. (a) Define following terms : 2

(i) Aspect Ratio

(ii) Resolution of an image

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(b) Find the composite transformation matrix for : 7

(i) Two successive rotations

(ii) Two successive scaling

(iii) Two successive translations

(iv) General pivot point rotation

(iv) General fixed point scaling

(c) Explain in detail Cohen Sutherland line clipping algorithm. 7

(d) Explain scan line polygon fill algorithm. 7

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Q. 2. (a) What do you mean by Beta Spline ? 2

(b) What are the various ways to represent a curve ? Derive a cubic bezier curve equation. 7

(c) Explain characteristics of B-Spline curves. Generate equations for blending functions for the cubic B-Splines considering d = 3. 7

(d) Construct the Bezier Curve of order 3 and with 4 polygon vertices A(1, 1), B(2, 3), C(4, 3) and D(6, 4) 15 7

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Q. 3. (a) What do you mean by vanishing point ? 2

(b) Explain Painter's algorithm of Back Phase removal with suitable example. 7

(c) Differentiate between 2D texture mapping & bump mapping. 7

(d) Explain Radiosity model in detail. 7

Q. 4. (a) What is the phenomenon of specular reflection ? 2

(b) Compare RGV, HVS and CYM colour models. 7

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(c) Distinguish between Gourand and Phong's method of shading. 7

(d) Explain and write steps of Depth Buffer algorithm. 7

Q. 5. (a) What do you mean by in betweens in animation ? 2

(b) What do you mean by the morphing in animation theory. 7

(c) Describe Grammar based model. 7

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(d) What are fractals ? Write down its basic characteristics, geometry & dimensions. 7

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